



UniUyo ID

Digital student ID cards with QR verification

The university already has a register of students with names and registration numbers. UniUyo ID turns each record into a digital ID card with a signed QR code. A lecturer scans the card and immediately sees whether the person is a UniUyo student, and whether they belong to the class being taught.

48

students in the register

7

courses, 64 registrations

36

past scans

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Demo logins: lecturer@uniuyo.edu.ng / lecturer123, student@uniuyo.edu.ng / student123, uduak.etuk@student.uniuyo.edu.ng / student123. Screens captured with the clock frozen at 2026-10-05.

1. Overview

Who	What they do
Student	Logs in to a wallet view of their ID card (front and back) with photo, registration number, department, level, expiry and QR code. Can show the QR full screen with a live clock, print the card, and see the classes it admits them to.
Lecturer	Opens the checkpoint for one of their classes and scans a card with the camera, uploads a photo of the QR, or types a registration number. Sees ADMIT, a warning or REJECT with the student's details and the three checks. Also sees each class register, who was admitted today, and a filterable scan log.
Anyone	Scanning the QR with a normal phone camera opens a public page that confirms the card is genuine and current, without showing course details.

The three questions a scan answers

1. Is this card genuine? (the QR signature is checked)
2. Is this a UniUyo student in good standing? (register, card version, suspension, expiry)
3. Does this student belong to this class? (course registration, then department)

Design

The layout follows the job. Lecturers get a checkpoint console: a class session strip with live counters, a camera viewport with a scan frame, and a large result panel that says ADMIT, NOT IN THIS CLASS or REJECT in colour with the three checks listed underneath. Students get a wallet view of a security-printed ID card that flips, prints and shows its QR full screen. The palette is UniUyo green with a gold accent, Inter for text and JetBrains Mono for registration numbers.

2. How the core logic works

Signed QR codes

```
QR text: https://<site>/verify/UU1.<base64url reg no>.<card version>.<signature>
signature = first 16 bytes of HMAC-SHA256(server secret, "UU1.<reg>.<version>")
```

Changing one character of the registration number or version breaks the signature, so a student cannot make a card for someone else. The signature is compared in constant time. When a card is reissued, `card_version` goes up and QR codes on the old card stop working.

Verdict rules (`src/lib/verify.ts`)

Check, in order	Result	Lecturer sees
Signature fails	invalid	Not a UniUyo ID
Reg. number not in register	unknown	Student not found
Card version is old	revoked	Old card
Status suspended	suspended	Student suspended
Expired or graduated	expired	ID card expired
Registered for the course	in_class	Belongs to CSC 311
Same department, not registered	same_department	Not registered for CSC 311
Different department	other_department	UniUyo student, not in this class

Every check is saved in the scans table with the lecturer, course and result, so the class list can show who was verified today. Lecturers can only scan for, and see rosters of, courses they teach.

Reading the QR

The camera scanner uses `getUserMedia` and the `jsQR` library to read frames in the browser. The same decoder reads uploaded photos. Typed registration numbers are normalised (upper case, single slashes) and looked up directly; the result says "Looked up by registration number" because no card was presented.

Dates

Rules take "today" as an argument. `UNIID_TODAY=YYYY-MM-DD` freezes the clock for demos and tests. Cards expiring within 30 days show a warning.

3. Architecture and data model

Browser

Server-rendered pages. Client components: checkpoint (camera, jsQR, result panel), ID wallet (flip, full-screen QR), login demo buttons.

Next.js 16

Server Components, a Server Action for each scan, QR SVGs rendered on the server, proxy.ts route guard, /api/health.

PostgreSQL

Drizzle ORM, SQL migrations from drizzle-kit.

Table	Columns	Rules
students	id, reg_no, first_name, last_name, other_name, gender, faculty, department, level, entry_year, status, id_issued_on, id_expires_on, card_version, blood_group	reg_no unique; the existing register
users	id, name, email, password_hash, role (student, lecturer), department, student_id	email unique; student_id links a login to a register entry
courses	id, code, title, department, level, units, venue, lecturer_id	code unique
enrollments	id, student_id, course_id	unique (student_id, course_id)
scans	id, lecturer_id, course_id, student_id, result, scanned_at	result is one of the eight verdicts

Route	Who	Purpose
/login	All	Log in, demo accounts
/card	Student	ID wallet, status and courses
/scan	Lecturer	Checkpoint for one class
/courses	Lecturer	Class overview
/courses/[id]	Lecturer	Attendance register
/log	Lecturer	Scan log with filters
/verify, /verify/[token]	Public	Card check for anyone
/api/health	Railway	Health check with database ping

4. Screen walkthrough

Sign in



Sign in

Pick a demo account or use your university email.

1

 Lecturer Dr. Aniefiok Ekong	 CSC student Ubong Akpan	 Law student Uduak Etuk
--	---------------------------------------	--------------------------------------

2

University email

lecturer@uniuyo.edu.ng

Password

.....

3

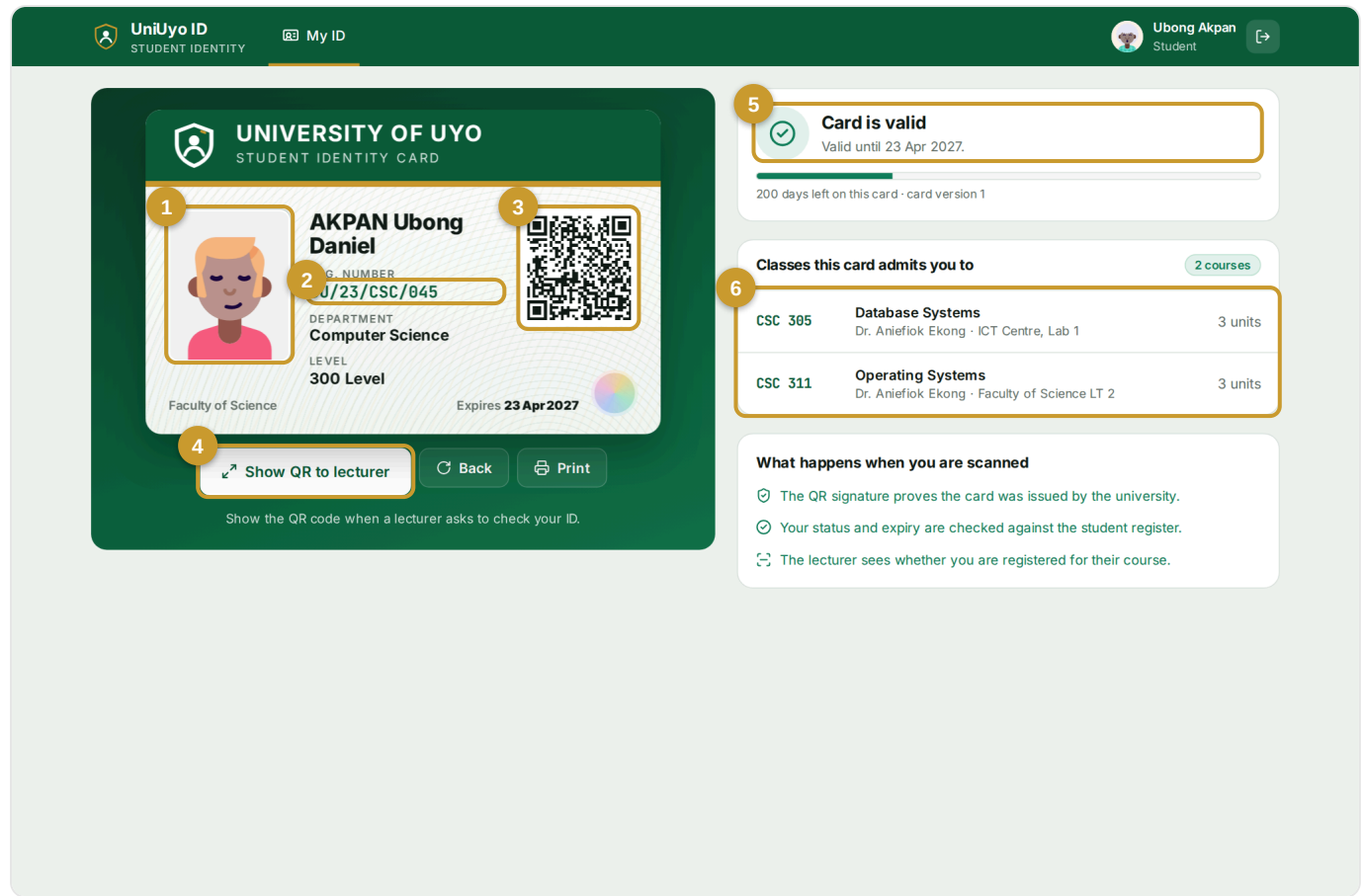
Sign in

Demo passwords: lecturer123 for the lecturer, student123 for every student.

Sign-in page.

- 1 Pick a demo account: the lecturer, a CSC 311 student, or a Law student for the 'not in this class' case.
- 2 Fields are prefilled with the lecturer account.
- 3 Sign in checks the bcrypt hash and sets a signed JWT in an HTTP-only cookie.
- 4 What the system does, over a real lecture theatre photo (Creative Commons).

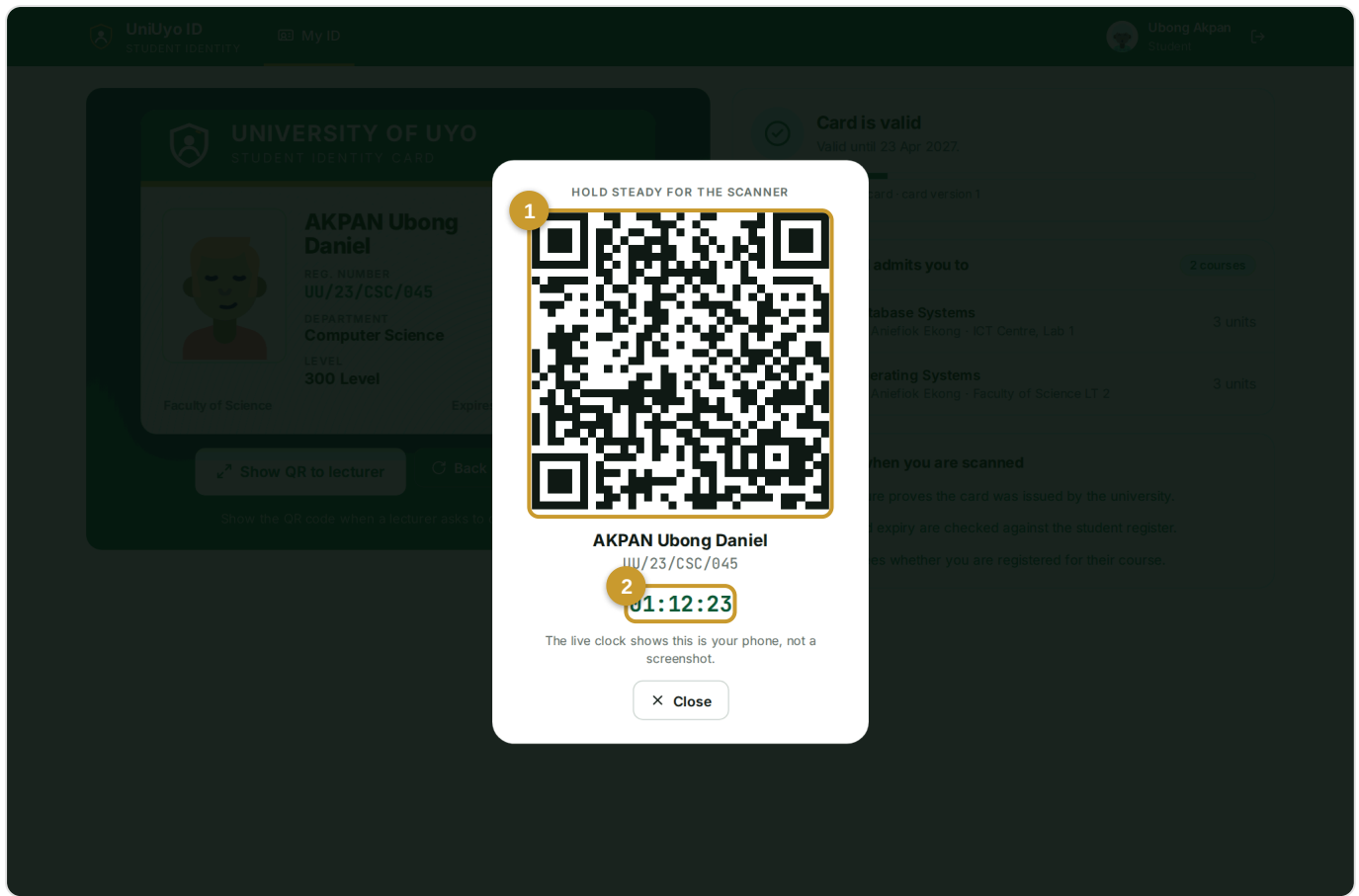
Student: ID wallet



The demo student's ID wallet.

- 1 ID photo: a generated portrait (DiceBear), never a real person's face.
- 2 Registration number, name, department and level from the student register.
- 3 Signed QR code. It links to the public verification page and cannot be forged without the server key.
- 4 Shows the QR full screen with a live clock, so a screenshot of someone else's card is easy to spot.
- 5 Card status from `cardState()`: valid, expiring within 30 days, expired or suspended.
- 6 The classes this card admits the student to.

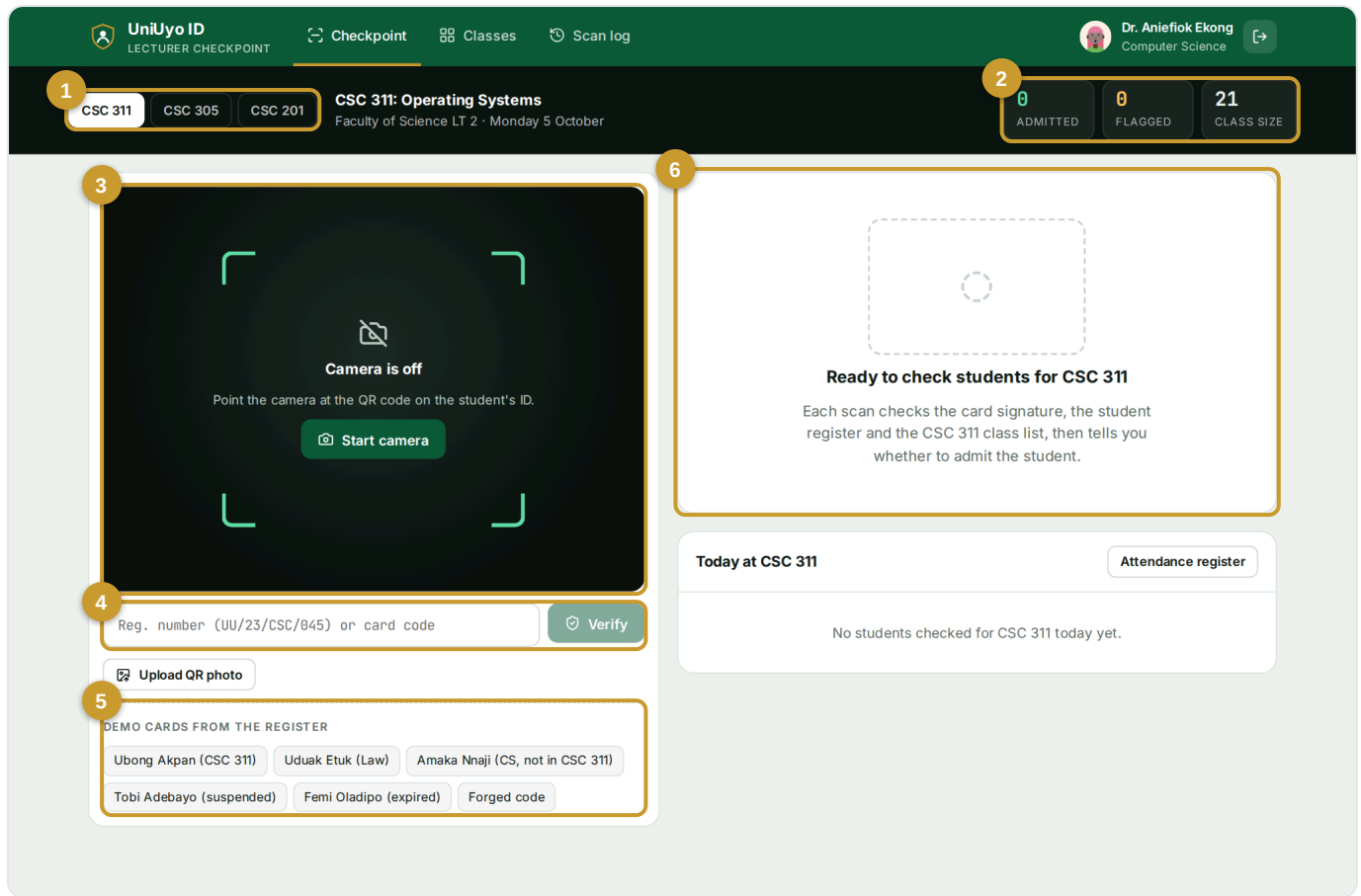
Student: QR shown full screen



The QR the lecturer scans.

- 1 Large QR for the lecturer's camera.
- 2 Live clock that ticks every second.

Lecturer: checkpoint



Checkpoint for CSC 311 before the first scan.

- 1 Switch between the lecturer's classes. Every check is made against the selected class.
- 2 Today's tally for the class: admitted, flagged and class size.
- 3 Camera viewport. jsQR reads the QR code from the video in the browser.
- 4 Type a registration number or paste a card code. Upload QR photo reads a picture instead.
- 5 Demo cards from the seeded register, for rehearsing with one device.
- 6 Result panel, waiting for the first card.

Lecturer: a student from another department

The screenshot shows the UniUyo ID Lector Checkpoint interface. The top navigation bar includes 'UniUyo ID LECTURER CHECKPOINT', 'Checkpoint', 'Classes', and 'Scan log'. The user is logged in as 'Dr. Aniefiok Ekong, Computer Science'. The current class is 'CSC 311: Operating Systems' in the Faculty of Science LT 2 on Monday 5 October. The class status shows 0 ADMITTED, 1 FLAGGED, and 21 CLASS SIZE.

The main interface is divided into two sections. The left section is a camera view with a 'Camera is off' message and a 'Start camera' button. Below the camera view is a 'Verify' button and a 'Upload QR photo' button. The right section displays a 'NOT IN THIS CLASS' alert for a student named ETUK Uduak, who is a Law student. The alert includes a student photo and details: REG. NUMBER UU/23/LAW/112, LEVEL 300, DEPARTMENT Law, and CARD EXPIRES 16 Jul 2027. Below the alert, there are three checks: 'Card is genuine' (QR signature matches), 'Student in good standing' (Active, card current), and 'Registered for this course' (Not on the class list). At the bottom, there is a 'Today at CSC 311' section with an 'Attendance register' button and a log entry for ETUK Uduak at 2:12 am, marked as 'Other department'.

1 NOT IN THIS CLASS
UniUyo student, not in this class

2

3

4

A Law student scanned at CSC 311.

- 1 The instruction in large type: NOT IN THIS CLASS, in amber.
- 2 Student photo and register details, so the lecturer can compare the face with the person.
- 3 The three checks behind the answer: card genuine, student in good standing, registered for this course.
- 4 Today's log for this class updates after every scan.

Lecturer: a registered student

The screenshot displays the UniUyo ID LECTURER CHECKPOINT interface. At the top, the header includes the UniUyo ID logo, navigation links for Checkpoint, Classes, and Scan log, and a user profile for Dr. Aniefiok Ekong, Computer Science. Below the header, a dark bar shows the selected class 'CSC 311: Operating Systems' and the date 'Monday 5 October'. On the right of this bar, three counters are visible: '2 ADMITTED', '1 FLAGGED', and '21 CLASS SIZE'. The main interface is divided into two panels. The left panel, labeled 'Camera is off', prompts the user to 'Point the camera at the QR code on the student's ID.' and features a 'Start camera' button. Below this, there is a search bar for 'Reg. number (UU/23/CSC/045) or card code' with a 'Verify' button, and an 'Upload QR photo' button. A section titled 'DEMO CARDS FROM THE REGISTER' lists several student cards, including 'Ubong Akpan (CSC 311)', 'Uduak Etuk (Law)', 'Amaka Nnaji (CS, not in CSC 311)', 'Tobi Adebayo (suspended)', 'Femi Oladipo (expired)', and 'Forged code'. The right panel, labeled '1 ADMIT', shows the student's profile for 'AKPAN Ubong Daniel', who is 'Registered for CSC 311 this semester.' It includes fields for 'REG. NUMBER' (UU/23/CSC/045), 'LEVEL' (300), 'DEPARTMENT' (Computer Science), and 'CARD EXPIRES' (23 Apr 2027). Below the profile, three status checks are listed: 'Card is genuine' (QR signature matches), 'Student in good standing' (Active, card current), and 'Registered for this course' (On the class list). At the bottom of this panel, it states 'CSC 311 - Checked from QR signature' and 'UniUyo student'. The bottom section of the right panel, titled 'Today at CSC 311', shows a list of students present at 2:12 am, including 'Ubong Akpan' (UU/23/CSC/045) marked 'In class' and 'Uduak Etuk' (UU/23/LAW/112) marked 'Other department'. An 'Attendance register' button is located at the top right of this section.

A CSC 311 student scanned.

- 1 Green ADMIT: genuine card, active student, registered for CSC 311.
- 2 The Admitted counter goes up.

Lecturer: classes

The screenshot shows the UniUyo ID Lector Checkpoint interface. At the top, there is a dark green header with the UniUyo ID logo and the text "LECTURER CHECKPOINT". Navigation links for "Checkpoint", "Classes", and "Scan log" are visible. The user's profile, "Dr. Aniefiok Ekong, Computer Science", is shown in the top right corner. The main content area is titled "THIS SEMESTER Your classes". It displays three class cards: "CSC 311 Operating Systems" (300 Level · 3 units), "CSC 305 Database Systems" (300 Level · 3 units), and "CSC 201 Computer Programming II" (200 Level · 3 units). Each card shows the "Admitted today" progress bar, the number of scans (e.g., "1 / 21" for CSC 311), and the "Last scan" time. A yellow callout box labeled "1" highlights the "CSC 311" card, and a yellow callout box labeled "2" highlights the "Open checkpoint" button on the same card.

Class ID	Class Name	Level	Units	Admitted Today	Last Scan	Open Checkpoint	Register
CSC 311	Operating Systems	300 Level	3 units	1 / 21	5 Oct, 2:12 am	Yes	Yes
CSC 305	Database Systems	300 Level	3 units	0 / 20	4 Oct, 9:09 am	Yes	Yes
CSC 201	Computer Programming II	200 Level	3 units	0 / 4	2 Oct, 9:07 am	Yes	Yes

Every class the lecturer teaches.

- 1 Each class with admitted-today progress, flags and the last scan time.
- 2 Opens the checkpoint for that class.

Lecturer: attendance register

UniUyo ID
LECTURER CHECKPOINT

Checkpoint

Classes

Scan log

Dr. Aniefiok Ekong
Computer Science

ATTENDANCE REGISTER · 2026-10-05

CSC 311 Operating Systems
Computer Science · 300 Level · Faculty of Science LT 2

Open checkpoint

1
21
Registered

1
Admitted today

20
Not yet checked

2
Card problems

Class list

Sorted by surname

STUDENT	REG. NUMBER	ID CARD	TODAY	LAST ADMITTED
 ABUBAKAR Halima	UU/23/CSC/078	Expiring	Not checked	Never
 ADEBAYO Tobi	UU/23/CSC/081	2 Suspended	Not checked	Never
 ADEYEMI Tunde	UU/23/CSC/056	3 Valid	Not checked	Never
 AKPAN Ubong Daniel	UU/23/CSC/045	Valid	Present	5 Oct, 2:12 am
 AKPANUDO Samuel	UU/23/EEE/310	Valid	Not checked	Never
 ARCHIBONG Eno	UU/23/CSC/040	Valid	Not checked	3 Oct, 9:12 am
 BASSEY Ekemini	UU/23/CSC/012	Valid	Not checked	26 Sept, 9:00 am
 EDET Ubong	UU/23/CSC/038	Valid	Not checked	3 Oct, 9:10 am
 EKANEM Emem	UU/23/CSC/034	Valid	Not checked	3 Oct, 9:08 am
 ESSIEN Ifiok	UU/23/CSC/021	Valid	Not checked	3 Oct, 9:00 am

CSC 311 register.

- 1 Registered, admitted today, not yet checked, and card problems.
- 2 A suspended student on the class list.
- 3 Present means admitted at the checkpoint today.

Lecturer: scan log

**UniUyo ID**
LECTURER CHECKPOINT

[Checkpoint](#)

[Classes](#)

[Scan log](#)

**Dr. Aniefiok Ekong**
Computer Science

EVERY CHECK YOU HAVE MADE

Scan log

1

All

Admitted

Flagged

Rejected

All courses

CSC 311

CSC 305

CSC 201

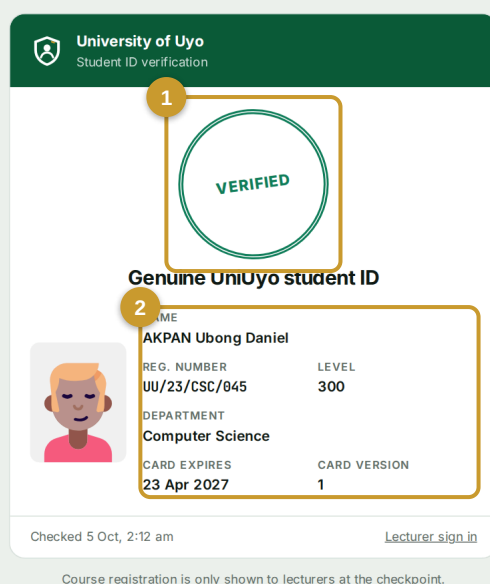
2

TIME	COURSE	STUDENT	DEPARTMENT	RESULT
5 Oct, 2:12 am	CSC 311	Ubong Akpan UU/23/CSC/045	Computer Science	In class
5 Oct, 2:12 am	CSC 311	Uduak Etuk UU/23/LAW/112	Law	Other department
4 Oct, 9:09 am	CSC 305	Femi Oladipo UU/21/CSC/003	Computer Science	Expired card
4 Oct, 9:06 am	CSC 305	Seun Ogunleye UU/23/CSC/074	Computer Science	Not registered
4 Oct, 9:03 am	CSC 305	Amaka Nnaji UU/23/CSC/088	Computer Science	In class
3 Oct, 9:22 am	CSC 311	Unrecognised code	-	Forged or invalid
3 Oct, 9:18 am	CSC 311	Tobi Adebayo UU/23/CSC/081	Computer Science	Suspended
3 Oct, 9:12 am	CSC 311	Eno Archibong UU/23/CSC/040	Computer Science	In class
3 Oct, 9:10 am	CSC 311	Ubong Edet UU/23/CSC/038	Computer Science	In class
3 Oct, 9:08 am	CSC 311	Emem Ekanem UU/23/CSC/034	Computer Science	In class
3 Oct, 9:06 am	CSC 311	Idara Umoh UU/23/CSC/031	Computer Science	In class

All checks, newest first.

- 1 Filter by result group (admitted, flagged, rejected) or by course.
- 2 Every check with time, course, student and result.

Public verification

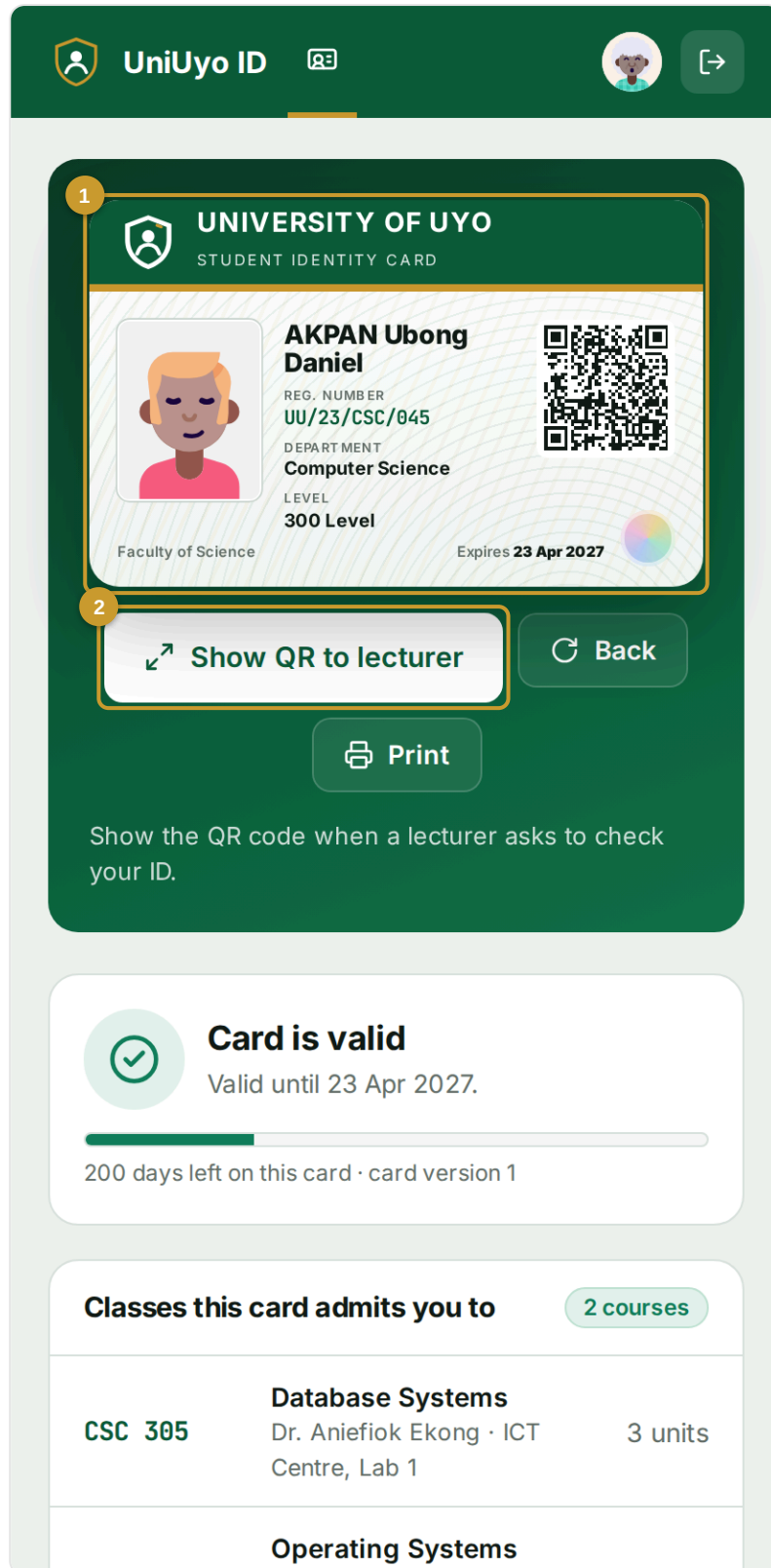


Opened from the QR code with any phone camera.

- 1 What any phone camera opens: a verification certificate with a stamp.
- 2 Only details printed on the card. Class registration needs a lecturer login.

5. Mobile view

All grids use `minmax(0, 1fr)`. The mobile e2e test fails if the page scrolls sideways, and checks that the result is scrolled into view.

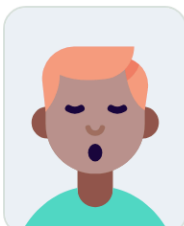


Student ID on a phone.

- 1** The card scales with container query units, so it keeps its shape on any phone.
- 2** Big button to show the QR full screen.

1


REJECT
Student suspended



ADEBAYO Tobi
The register shows this student as suspended.

REG. NUMBER
UU/23/CSC/081

LEVEL
300

DEPARTMENT
Computer Science

CARD EXPIRES
02 Jan 2028

2

☒ **Card is genuine** · QR signature matches
☒ **Student in good standing** · Suspended
☐ **Registered for this course** · Not checked

CSC 311 · Checked from QR signature UniUyo student

Today at CSC 311
Attendance register

2:12 am	 Tobi Adebayo UU/23/CSC/081	Suspended
2:12 am	 Ubong Akpan UU/23/CSC/045	In class
2:12 am	 Uduak Etuk UU/23/LAW/112	Other department

Lecturer checkpoint on a phone after a scan.

- 1 On phones the page scrolls to the result after each scan: REJECT, student suspended.
- 2 Which check failed.

6. Running locally

```
service postgresql start
sudo -u postgres createdb uniid
sudo -u postgres createdb uniid_test
cp .env.example .env      # DATABASE_URL, SESSION_SECRET
npm install
npm run db:migrate
npm run db:seed
npm run dev                # http://localhost:3000
```

Script	What it does
dev / build / start	Development server, production build, production server
start:prod	Migrate, seed only when empty, next start -H 0.0.0.0 (Railway)
db:generate, db:migrate, db:seed	New migration, apply, reset and seed
test:unit, test:integration, test:e2e	Test suites
docs:pdf	Rebuild this PDF

The camera needs HTTPS or localhost. On the deployed site it works on phones and laptops.

7. Testing

19

unit tests passed

14

integration tests passed

8

e2e tests passed

Suite	Covers
Unit	Token format, round trip, edited reg number, bumped version, wrong key, junk input, URL extraction; every verdict and their order; card states; reg number normalising; dates; sessions.
Integration (Postgres)	Seed size; in class; full URL input; Law student; same department; suspended, expired, old and new card versions; forged and unknown; typed reg number; lecturer cannot scan for other courses; course and scan scoping; roster counts; student record scoping; public lookup; scan log.
End to end (Playwright)	Login prefill; student card, QR link and public page; lecturer scans a class member, a Law student and a forged code; QR photo upload decoded in the browser; inline error and class list; students blocked from lecturer pages; mobile card and mobile scan.

Latest run: unit 19, integration 14, 0 failed; e2e 8 of 8 passed (7 desktop, 1 mobile).

8. Deployment

Runs on Railway in the **school-projects** project as the **uni-digital-id** service with its own **uni-digital-id-postgres** database, deployed from the main branch of the GitHub repository.

Setting	Value
Build	npm run build
Start	npm run start:prod
Health check	/api/health (503 if Postgres is down)
DATABASE_URL	\${{uni-digital-id-postgres.DATABASE_URL}}
SESSION_SECRET	random value; also the key for QR signatures
NODE_ENV	production

9. Five minute presentation script

Time	What to do and say
0:00	Problem: lecturers cannot easily tell whether someone in class is a UniUyo student, or whether they belong to the course. Paper IDs are easy to fake. We use the existing student register and add a signed QR code.
0:30	On a phone, log in as the student (Ubong Akpan). Show the wallet: the card, Back to flip it, the classes it admits them to. Tap Show QR to lecturer; point out the live clock.
1:15	On the laptop, log in as the lecturer. The checkpoint opens for CSC 311 with today's counters.
1:40	Click Start camera and hold the phone's QR up: green ADMIT, the student's photo and the three checks. The Admitted counter goes up.
2:20	Type the Law student's registration number UU/23/LAW/112: amber, UniUyo student, not in this class. This is the case the lecturer asked for.
2:50	Click the forged code sample: red, Not a UniUyo ID. Explain the HMAC signature in one sentence. Click the suspended sample: red.
3:30	Open Attendance register: present today, not yet checked, card problems. Then Scan log, filtered to Rejected.
4:00	Scan the student's QR with the phone's normal camera app: the public page says Genuine UniUyo student ID.
4:20	Architecture and tests: Next.js Server Components and Actions, Drizzle with Postgres, pure verdict functions, test counts, Railway with health checks.
4:45	Next steps: attendance export per lecture, card reissue workflow at the ICT Centre. Questions.

Image credits

Campus and lecture photos are Creative Commons images found through Openverse, stored in `public/images` with credits in `public/images/credits.json`. Student photos are generated portraits (DiceBear), not real people.